

OASIS Value Proposition for Chandarana Foodplus

The Problem We're Solving

"You know when a shelf goes empty on a Saturday afternoon. What you don't see is the KES 50,000 walking out the door every time."

What Was Learned from Data

We have analyzed a flagship store. Here's what we found:

The Current State: "Low fill rate"

Metric	What We Measured	The Gap
SKU Coverage	23,000+ barcodes in GRN database	Only ~40% get consistent reorders
Order Pattern	Average 3.2 days between supplier orders	Many suppliers go 14+ days without a PO
Fill Rate (Estimated)	~65-70%	30% of customer demand hits an empty shelf
Dead Stock Ratio	~18% of capital	Over KES 15M locked in zero-velocity SKUs
Fast Mover Stockouts	12-15 stockout days/month (Rice, Sugar, Oil)	Your anchors are failing you

The Hidden Cost

Based on your sales data, every 1% improvement in fill rate = ~KES 2.8M additional annual revenue for a Mega store.

OASIS PLATFORM		
1. STOCK ALLOCATION MODEL	2. OASIS ORDER ENGINE (Replenishment)	3. SIMULATION ENGINE
"What do I buy on Day 1?"	"When do I reorder?"	"What happens if...?"
Greenfield allocations for new stores or expansions	Dynamic ROP calculations for running stores	Monte Carlo risk modeling before you commit capital

- + BONUS: Product Scorecard Generator
(Automated velocity tiering, supplier scoring, department mapping)

System 1: Stock Allocation Model

What It Does

Takes your opening capital and tells you exactly what to buy, how many units, and from which suppliers—all in one Excel output a clerk can action.

The Intelligence Inside

Multi-Pass Waterfall Logic

- PASS 1: WIDTH ("Make the shelf look full")
- Essential SKUs get 1 pack minimum
 - Price ceiling filters out luxury items from small stores
 - Dead stock (D-Class) automatically excluded
- PASS 2: DEPTH ("Stock what sells")
- Remaining budget flows to velocity winners
 - 80/20 split: 80% to staples, 20% to discretionary
 - Wallet constraints prevent any one department from hoarding
- PASS 2B: OPTIMIZER ("Use every shilling")
- Unused department budgets redistribute to staple anchors
 - Targets 100% budget utilization

Dynamic Store Profiling

Your dukas and your Yaya behave differently. Our logic knows:

Store Tier	Budget	Depth Days	Price Cap	Staple Priority
Micro (Duka)	<200K	7 days	KES 300	95% Fast Five
Mini-Mart	1M	14 days	KES 2,500	80% Staples
Supermarket	10M	21 days	KES 20,000	60% Staples
Mega	50M+	30 days	No cap	Balanced

What This Replaces

Current Process	OASIS Allocation
HQ Order Frequency + Branch local orders	Data-driven scorecard with 12M+ GRN records
Supplier reps push slow movers	Velocity-tiered ranking (A/B/C/D Class)
Over / Under Orders	Pack-rounded orders that hit exact capital
Bulk buys that don't fit	Max pack limits prevent overbuy
New product paralysis	Lookalike demand for untested SKUs

System 2: OASIS Order Engine (Replenishment)

What It Does

Runs every day (or on your frequency). Scans live stock, calculates what's below Reorder Point, and generates a PO draft you can approve or edit.

The Intelligence Inside

Dynamic Reorder Point Formula

$ROP = (\text{Lead Time} + \text{Safety Days}) \times \text{Daily Velocity} \times \text{Month Factor}$

Where:

- Lead Time = From your GRN data (actual supplier performance)
- Safety Days = 3-day buffer (adjustable by category)
- Daily Velocity = Average of last 30 days (smoothed by CV)
- Month Factor = Seasonality index (January ≠ September)
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Risk-Adjusted Ordering

Risk Trigger	Buffer Applied
Supplier reliability < 70%	+25% depth
Demand CV > 0.8 (volatile item)	+15% depth
Lead time > 14 days	+2 safety days
Fresh item (Daily supplier)	Max 2-day stock

What This Replaces

Current Process	OASIS Replenishment
"We're out of Jogoo—order more!"	Proactive reorder 5 days before stockout
Oversized emergency orders	Right-sized based on actual velocity
Forgotten slow-movers pile up	D-Class items blocked until manually approved
Supplier minimums cause bulk buys	MOV-aware batching across SKUs

System 3: Simulation Engine

What It Does

Before you commit capital, run a 30/60/90-day simulation. See projected fill rate, stockout days, revenue capture, and cash flow.

The Intelligence Inside

Monte Carlo Sales Model

Daily Demand = Base Velocity × Day Factor × Month Factor × Store Scale

Where:

- Day Factor = {Mon: 0.9, Sat: 1.3, Sun: 0.8} # Weekend spikes
- Month Factor = Seasonality from your historical sales
- Store Scale = Budget-proportional adjustment

What You Can Test

Scenario	What It Shows
"What if I only have KES 3M for this branch?"	Exact fill rate at that budget
"What if my flour supplier delays 4 days?"	Revenue loss from stockouts
"What if December demand is 2x normal?"	Whether your safety stock holds
"What if I consolidate to 3 rice suppliers?"	Impact on depth vs. variety

Simulation Outputs

- ****Fill Rate Curve**** (Day-by-day %)
 - ****Stockout Report**** (Which SKUs, how many lost sales)
 - ****Cash Flow Table**** (Orders, Lead Times, Arrivals)
 - ****Supplier Concentration Risk**** (HHI Index)
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What the Numbers Show

Simulation Results: KES 3.5M Mini-Mart (30 Days)

Metric	Before (Estimated)	After (OASIS Simulation)
Fill Rate	55%	**98.2%**
Stockout Instances	150+	**3**
Revenue Captured	KES 4.2M	**KES 7.8M**
Dead Stock (Day 30)	35% of capital	**8%**
Emergency Orders	6	**0**
Supplier Order Efficiency	Random	**MOV-optimized batches**

*****Net Uplift: +85% revenue capture from same capital base.*****

Why This Matters for Chandarana

The Strategic Value

1. **Branch Opening Velocity**

Open new stores faster. Instead of 3 weeks of "ordering experiments," deploy OASIS allocation on Day -7.

2. **Capital Efficiency**

Your KES 3M works like KES 5M. Shelf presence + depth without overstocking.

3. **Supplier Consolidation**

For small formats, we automatically consolidate to Top 3-5 suppliers per department. Fewer invoices, better terms, simpler logistics.

4. **Category Manager Visibility**

Every allocation comes with a logic trace: "PASS 1: WIDTH → MDQ 3 → 6 Units [STAPLE ANCHOR]". Full transparency.

5. **Risk Modeling Before Commitment**

Run 10 scenarios for Syokimau. Know which capital level hits 95% fill before you wire money.

Pilot Proposal

Success Metrics

Metric	Target
Fill Rate (Anchor SKUs)	>95%
Order-to-GRN Match	>90% (recommendations actioned)
Emergency Orders	-50% vs. control period
Dead Stock Reduction	-30% vs. opening baseline

Investment & ROI

Break-Even Analysis

Store Type	Capital	Estimated Lift	Break-Even
Mini-Mart	KES 3.5M	+15% revenue	~45 days
Supermarket	KES 10M	+12% revenue	~30 days
Mega	KES 50M+	+8% revenue	<21 days

Next Steps

1. Be able to consolidate data for all Branches – Will help us make our engine prefer internal transfers to orders , control dead stock automatically through Branches
2. Integration of Promotional calendars and offers into logic : to explain some demand spikes
3. Integration of actual size metrics of stores to be able to determine how many facing SKU's can be on the shelves at the same time
4. **Key Validation Points for Anil Thakkar**
 1. **Demand is REAL** - Simulation uses your actual 24K SKU sales velocity from Chandarana cash registers
 2. **Velocity Tiering Matches** - Your 5.5% A-Class SKUs are prioritized for depth
 3. **Supplier Lead Times are ACTUAL** - Calculated from real GRN gaps (Brookside=1 day, Del Monte=9 days)
 4. **The Fill Rate Gap is your ROI** - 65% → 97% = ~**KES 25-30M/year** in captured revenue
 5. **Inventory Turns Improvement** - 2.5x → 9.5x = better cash flow, less expiry

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Intelligent Retail Replenishment